

Phase-1

- Load dataset
- Check the number of rows and columns
- Check datatypes of the columns
- Select the numeric columns
- Do train-test partition with 80:20 ratio (random_state=23)
- Build KNN classifier with K=15 and report accuracy, precision, recall, f_measure
- Display confusion matrix

Phase-2

- Find optimal/best value of K using accuracy elbow plot
 - K=1, Accuracy=67.5%
 - But should we choose this K?
- But for credit default prediction recall is more important
- Secondly, we have an unbalanced dataset with maximum records for non-default and hence accuracy measure should not be sole our criteria for model evaluation
- Find optimal/best value of K using recall elbow plot
 - K=1, Recall=42.65%
 - Should we choose this K?

Phase-3

- Show confusion matrix with $K=1$
- Let's understand Receiver Operating Characteristics (ROC) and Area Under Curve (AUC)
- ROC: TPR vs FPR, captures the discriminating power of the model
- Check ROC for the model with $K=1$
 - AUC: 61.5 not very good
 - $K=1$ will result in overfit model
- Change `random_state=544` and rerun the previous steps and observe the change of results!!!!
- So, why this happens and what should we do?

Phase-4

- Reset the random_state=23
- Run grid search with cross validation with number of folds=10
- Note the parameters of the best model
- Build the KNN classifier with best K=21
- Run on test dataset
- Do you see any improvement?
- Now change the random_state=544 and rerun the cross-validation steps
- What changes do you notice?

Phase-5

- We can also run cross validation using precision and recall as evaluation metrics
- For that, we need to label encode the outcome variable
- This is also shown in the worked-out example

Phase-6

- So, we have hit the wall!
- Not much improvement with cross validation
- Need to include other variables in the model
- First include any two categorical features along with the numeric features
- Note, you must convert the categorical features using one-hot encoding before you can run KNN

Phase-6 (contd.)

- Run grid search with cross-validation and find best parameters
- Build model with the best parameters
- Check accuracy, precision, recall, f_measure and ROC
- Do you see any improvement?
- What will be your conclusion?

Phase-7

- Compare your best KNN model with Logistic regression model
- Which one would you prefer?
- Why?

Phase-8

- Now we have a KNN model with all independent variables
- Is it necessary to keep each independent variable in the model?
- We can plot **Feature Importance**
- Based on feature importance score and domain knowledge, we can do **Feature Selection**
- Feature selection helps to simplify the model and achieve comparable performance